Power Engineering Lab — Smart Grid Research & Simulation Platform —
Portland State University

Smart Water Heater Test Station

Hardware Design, Station Build & System Integration

Embedded System Hardware for Control, Data Acquisition, and AC Power Monitoring
for Smart-Grid Water Heater Evaluation

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Document Purpose. This report documents the complete design, build, and validation of the Smart Water Heater Test Station at the Power Engineering Lab, Portland State University. It covers the full project lifecycle from system architecture and PCB hardware design through electrical installation, sensor integration, and system-level testing. It is intended as the primary technical reference for Prof. Robert Bass, lab researchers, and future students working with this platform.

Questions and Support. For questions about the hardware design, PCB, wiring, or software, contact the system designer: **Nawaf Altalhi** (Undergraduate Research Assistant, Power Engineering Lab, Portland State University). Refer also to the Power Lab documentation repository (GitHub / Google Drive) for schematics, software, and the wall-mounted wiring guide (Doc WCG-1, Rev A).

Associated Project Files. Supporting files for this project include the EasyEDA schematic and PCB source files, Gerber fabrication files, schematic exports, wiring guide, software scripts, test data, and station photos. These files are organized in the project documentation repository for future reference, maintenance, and revision work.

Smart Water Heater Test Station installed in the Power Engineering Lab, Portland State University. The platform integrates four electric water heater units, custom PCB-based monitoring and control hardware, AC distribution panels, industrial sensors, and Raspberry Pi–based data acquisition for smart-grid research.

1 Project Overview

The Smart Water Heater Test Station is an embedded industrial monitoring and control platform developed at the Power Engineering Lab at Portland State University. The station was designed and built to support automated testing and long-term evaluation of residential electric water heaters (EWHs) in a smart-grid research environment.

The platform integrates four EWH units from different manufacturers, each paired with a dedicated Raspberry Pi controller and a custom-designed PCB. The system enables real-time monitoring of water temperature, flow rate, AC voltage, AC current, power consumption, and energy behavior. Water flow is controlled automatically through relay-actuated solenoid valves. Communication with each water heater uses the RS-485/ANSI CTA-2045 protocol, which allows the controller to issue grid-service commands and collect operational responses.

1.1 Scope of Work

All hardware design, electrical installation, procurement, integration, and validation work described in this document was performed by Nawaf Altalhi:

- System architecture definition: identifying monitoring and control requirements, selecting the 4–20 mA industrial sensor standard, and defining the communication strategy (RS-485 / CTA-2045)
- Component selection and evaluation: flow transmitter, temperature transmitters, solenoid valve, ADC, power monitoring IC, relay, and all supporting circuitry
- Full component sourcing and procurement (see Appendix B)
- Hardware and circuit design: schematic capture of all subsystems including analog front-end, data acquisition, relay driver, RS-485 interface, and AC power monitoring
- PCB layout design (EasyEDA): 2-layer board with dedicated ground planes on both top and bottom copper layers across the low-voltage domain; the high-voltage section is physically isolated by board-level cutouts and enforced creepage distance, with no shared ground plane between the two domains; four units fabricated and assembled
- Single-unit bench testing: verification of PCB sensor inputs, ADC communication, relay switching, and power monitoring before full station installation
- Station planning and physical layout: each unit includes a dedicated pipe assembly installed in sequence — hot water temperature transmitter, solenoid valve, manual flow-adjustment valve, flow transmitter — terminating at the sink
- Main enclosure selection and AC panel build: two Hammond N1J884 NEMA 1 panels,

each housing two Eaton FAZ-B25-2-NA-L 2-pole 25 A breakers on DIN rail — four breakers total, one per water heater

- Full AC wiring inside the main enclosure: L1, L2, Neutral, and Ground distribution using terminal blocks
- Per-unit IP65 enclosure selection, layout, and build (one per water heater): each enclosure houses the custom PCB, Raspberry Pi, and cooling fans, with cable grommet entries for AC lines, sensor cables, and 24 VDC power supply
- Sensor and plumbing installation: temperature transmitters, solenoid valves, flow adjustment valve, and flow transmitter per pipe order
- Real-time data acquisition: water draw and all sensor readings collected through the MAX1238 12-bit ADC over I^2C and processed by the Raspberry Pi
- Python data-logging software running on each Raspberry Pi
- Full system testing, ACS37800 power-monitoring calibration, and validation
- Production of wall-mounted wiring and connection guide (Doc: WCG-1, Rev A) for lab technicians and future students

1.2 Station Configuration Summary

Table 1: Water heater unit configuration.

Unit	Panel	Breaker Pos.	AC Configuration	Notes
WH1	P1	Pos. 1	240 VAC (L1–L2)	Standard
WH2	P1	Pos. 3	120 VAC (L1–N)	Neutral used in place of L2
WH3	P2	Pos. 2	240 VAC (L1–L2)	Standard
WH4	P2	Pos. 4	240 VAC (L1–L2)	Standard

Panel arrangement: Panel 1 feeds WH1 and WH2. Panel 2 feeds WH3 and WH4. Main breakers are FAB-NDP1P-A (1,3) for Panel 1 and FAB-NDP1P-A (2,4) for Panel 2.

Table 2: High-level system parameters.

Parameter	Value
Number of water heater units	4
Controller per unit	Raspberry Pi
Custom PCBs designed and built	4 (one per unit)
Sensor protocol	4–20 mA industrial current loop
WH communication	RS-485 / ANSI CTA-2045
Power monitoring IC	ACS37800KMACTR-090B3
ADC	MAX1238, 12-bit, I^2C
Flow control	Relay-actuated solenoid valve, 120 VAC
Panel breakers	Eaton FAZ-B25-2-NA-L, 25 A, 2-pole
Main panel enclosures	Hammond N1J884, NEMA 1, steel, wall-mount (×2)
Per-unit enclosure	IP65 ABS, $11 \times 7.4 \times 5.5$ in (×4)
Sensor supply	24 VDC, 1.5 A (one adapter per unit)

2 System Architecture

Figure 1 shows the full embedded monitoring and control architecture for the station. Each of the four units operates independently with its own Raspberry Pi, custom PCB, sensors, and water heater connection. The PCB consolidates all sensor interfacing, signal conditioning, data acquisition, valve control, power monitoring, and RS-485 bridging to the water heater.

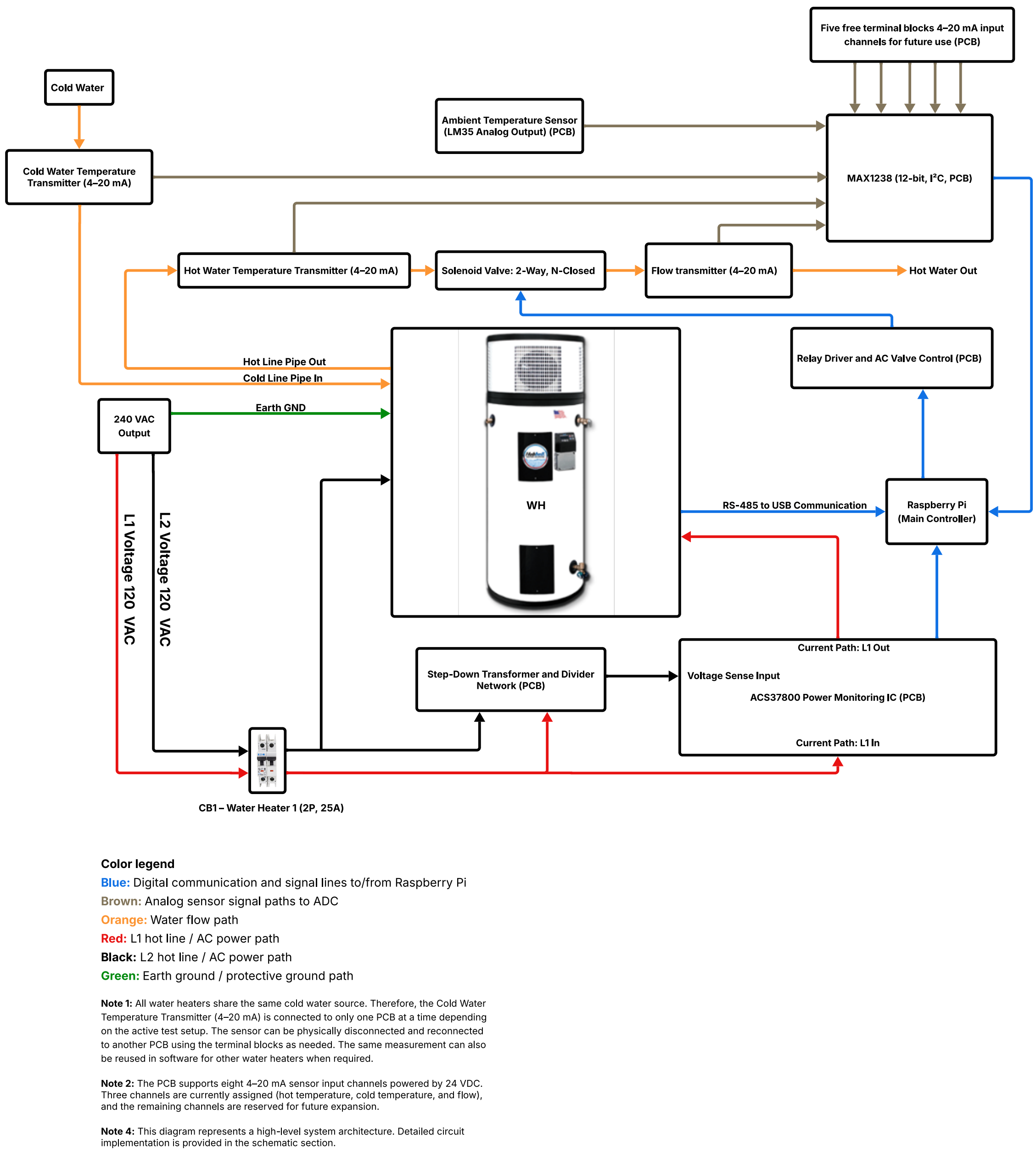

Figure 1: Embedded monitoring and control architecture for the Smart Water Heater Test Station. Each of the four units consists of a Raspberry Pi, a custom PCB, industrial 4–20 mA sensors, RS-485 communication to the water heater, and a relay-driven solenoid valve. All four units share the same cold water supply line.

2.1 Functional Subsystems per Unit

- **Temperature monitoring.** Hot water temperature is measured by a TWTADE 4–20 mA transmitter installed on the hot water pipe of each water heater. Cold water temperature is measured by a single shared transmitter on the common cold supply line; it is physically connected to one PCB at a time and the value is shared in software as needed.
- **Ambient temperature.** An LM35 analog sensor is mounted directly on the PCB and reads ambient temperature inside the enclosure. It is not part of the pipe assembly.
- **Flow monitoring.** A variable-area 4–20 mA flow transmitter (0.2–10 GPM, 3/4 in NPT) measures the water draw rate. A manual lever-handle flow-adjustment valve installed in series allows repeatable flow setting for controlled tests.
- **Flow control.** A 2-way normally closed solenoid valve (120 VAC) is switched by the relay on the PCB, driven from Raspberry Pi GPIO17. The valve is installed upstream of the flow transmitter in the pipe order.
- **AC power monitoring.** The ACS37800 IC measures RMS voltage, RMS current, power factor, active power, reactive power, and energy from the water heater supply lines. Data is read over I^{2}C by the Raspberry Pi.
- **Data acquisition.** All analog sensor signals are digitized by the MAX1238 12-bit ADC over I^{2}C and processed by the Raspberry Pi.
- **WH communication.** A USB–RS-485 adapter connects the Raspberry Pi to the water heater CTA-2045 port for grid-command exchange and operational data collection.
- **Data logging.** A Python script on each Raspberry Pi reads all sensor channels and power data continuously and logs to a local CSV file.

3 Hardware Design

Four identical PCBs were designed, fabricated, assembled, and installed — one per water heater unit. The PCB consolidates all sensing, signal conditioning, data acquisition, control, and power-monitoring functions into a single board that connects directly to the Raspberry Pi 40-pin GPIO header.

3.1 Schematic Overview

The PCB schematic is organized into eleven functional blocks as labeled in Figure 2:

- **Temperature & Flow Transmitters — 4–20 mA Sensor Interface:** eight sensor input channels; three are active in the current station configuration (hot water temperature, cold water temperature, and flow transmitter) and five are available for future use or additional sensors; each channel uses two 240 Ω resistors in parallel for current-to-voltage conversion

- **Op-Amp Signal Conditioning:** two LM324 quad op-amp stages configured as unity-gain buffers, isolating the sensor signals before the ADC input
- **MAX1238 ADC Interface — 12-bit I^2C Analog Front End:** digitizes all conditioned analog signals on channels AIN0 through AIN7
- **Raspberry Pi GPIO Header:** 40-pin interface carrying I^2C, GPIO control signals, and the 3.3 V and 5 V power rails to the PCB
- **I^2C Level Shifter — 3.3 V ↔ 5 V:** TXS0104EQPWRQ1 bridging the Raspberry Pi 3.3 V logic and the MAX1238 5 V logic
- **24 VDC Power Input:** locking barrel jack with 100 µF bulk and 0.1 µF decoupling capacitors supplying the sensor interface rail
- **Relay Driver for AC Valve Control:** MMBT3904 transistor, G5LE-1A4 relay, 1N4007W flyback diode, and dual status LEDs controlled from GPIO17
- **Step-Down Transformer and Divider Network:** BV302S12015 transformer with MOV surge protection and resistor divider scaling the AC voltage to the ACS37800 VINP input
- **AC Power Monitoring & Alert Interface — ACS37800:** measures VRMS, IRMS, power factor, active and reactive power over I^2C; DIO_0 and DIO_1 connected to GPIO18 and GPIO26 for overcurrent alert and status interrupt
- **RS-485 Communication Interface:** two terminal blocks connecting the USB–RS-485 adapter on the Raspberry Pi to the water heater CTA-2045 port
- **Temperature Sensor — LM35 Analog Output:** PCB-mounted ambient temperature sensor connected directly to ADC channel AIN4

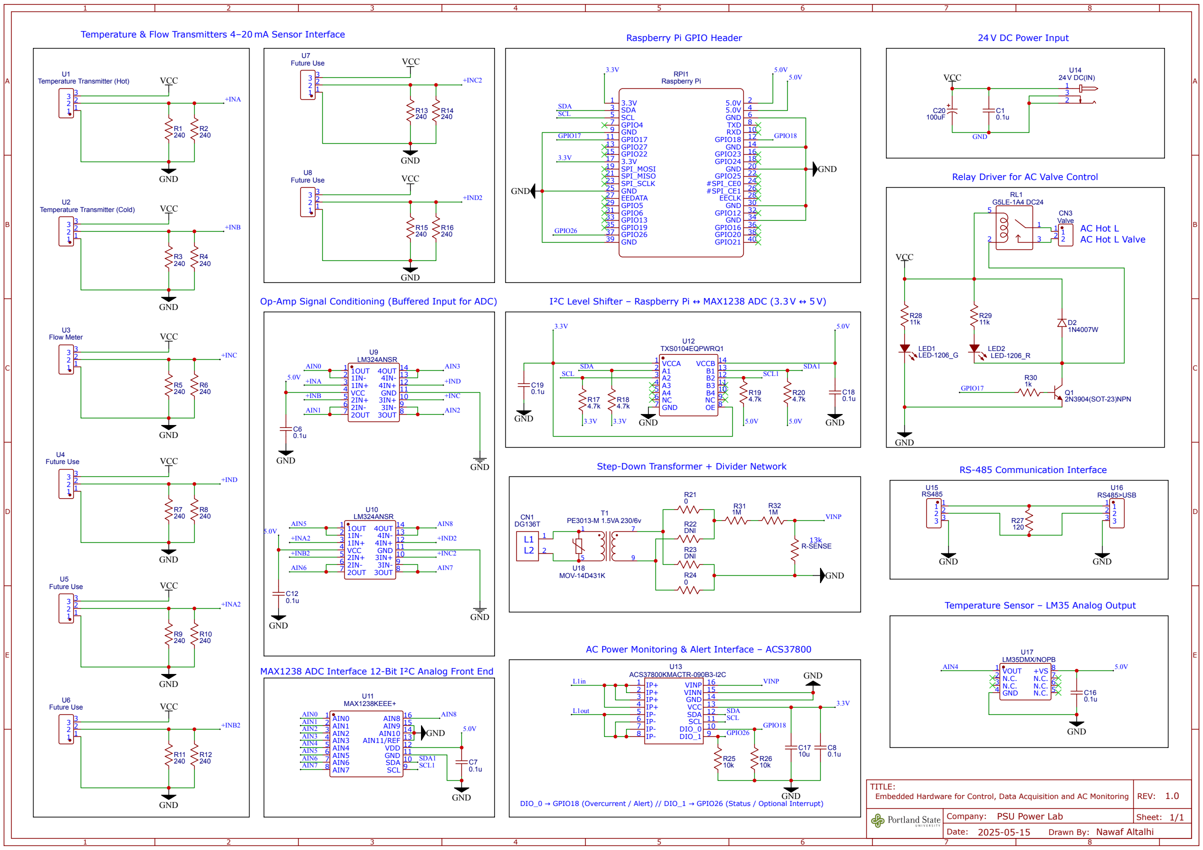

Figure 2: Full PCB schematic — Embedded Hardware for Control, Data Acquisition and AC Monitoring (Rev 1.0, EasyEDA, PSU Power Lab). Eleven functional blocks are labeled on the schematic. Detailed circuit description for each block is given in Sections 3.3–3.7.

3.2 Power Architecture

The PCB operates across three isolated voltage domains:

Table 3: PCB voltage domain summary.

Domain	Source	Powered Circuits
24 VDC	External wall adapter	All eight 4–20 mA sensor terminal blocks
5 V	Raspberry Pi rail	LM324 op-amps, MAX1238 ADC, LM35 sensor, 5 V side of I^2C level shifter
3.3 V	Raspberry Pi rail	ACS37800 power-monitoring IC, 3.3 V side of I^2C level shifter

The 24 VDC input arrives through a locking barrel jack (BKZ series). Input filtering uses

a 100 µF electrolytic capacitor (C20) for bulk decoupling and a 0.1 µF ceramic capacitor (C1) for high-frequency noise rejection. The Raspberry Pi is powered by its own external adapter and does not receive power from the PCB.

3.3 Sensor Interface and Signal Conditioning

The PCB provides eight 4–20 mA input channels powered from the 24 VDC rail. Three are active; five are reserved for future expansion.

Each 4–20 mA current is converted to a voltage using two 240 Ω resistors in parallel (120 Ω equivalent), producing 0.48 V at 4 mA and 2.40 V at 20 mA. The resulting voltage is buffered by an LM324 op-amp stage (unity gain) before the ADC, isolating the sensor loop from ADC input capacitance. Decoupling capacitors (0.1 µF) are placed at all op-amp supply pins.

The LM35 ambient sensor is connected directly to ADC channel AIN4, bypassing the op-amp stage. It produces 10 mV/°C and requires no signal conditioning.

3.4 Data Acquisition and I^{2}C Level Shifting

Analog signals are digitized by the MAX1238 12-bit ADC operating at 5 V. Channels AIN0 through AIN7 receive conditioned sensor outputs. A TXS0104EQPWRQ1 bidirectional I^{2}C level shifter bridges the 5 V ADC and the 3.3 V Raspberry Pi. Pull-up resistors of 4.7 kΩ are placed on each side. A 0.1 µF decoupling capacitor is placed at the MAX1238 VDD pin and at both level-shifter supply rails.

3.5 Relay Driver and Valve Control

GPIO17 drives the base of transistor Q1 (MMBT3904, NPN SOT-23) through R30 (1 kΩ). Q1 switches relay RL1 (G5LE-1A4, 24 V coil, SPST 10 A), which connects the 120 VAC line to the solenoid valve through connector CN3. Flyback diode D2 (1N4007W, SOD-123FL) suppresses the inductive spike at relay turn-off, protecting Q1 and the GPIO. LED1 (green) and LED2 (red) indicate valve state as described in Doc WCG-1.

3.6 RS-485 Communication Interface

RS-485 lines terminate at PCB screw terminals and connect via a USB–RS-485 adapter to the Raspberry Pi. Terminal and wire assignments are given in Doc WCG-1. This path carries CTA-2045 protocol messages between the Raspberry Pi and the water heater control port.

3.7 AC Power Monitoring

The power-monitoring section is based on the ACS37800KMACTR-090B3-I2C.

Voltage sensing. L1 and L2 enter through connector CN1. A varistor (MOV-14D431K, 430 V, 15.5 mm disc) protects the transformer primary against surges. Transformer T1 (BV302S12015, 1.5 VA, 230 V primary) steps down the AC voltage. Under no-load conditions the secondary measured approximately 20 VAC. A resistor network (R31 = 1 MΩ, R32 = 1 MΩ, R-SENSE = 13 kΩ) scales this to the ACS37800 VINP input, verified below the 250 mV peak limit (see Section 6).

Current sensing. L1 passes directly through the ACS37800 internal current path (IP+ to IP−). No external shunt is required.

Interface and alerts. The IC is powered from 3.3 V (C17 = 10 µF, C8 = 0.1 µF). Two digital outputs connect to the Raspberry Pi: DIO_0 → GPIO18 (overcurrent/alert, R25 = 10 kΩ), DIO_1 → GPIO26 (status/interrupt, R26 = 10 kΩ).

4 PCB Layout

4.1 Board Overview and Mechanical Design

The PCB measures 122.7 mm × 90.2 mm on a 2-layer board. Figure 3 shows the top copper layer and Figure 4 shows the bottom copper layer. The board follows a strict left-to-right partitioning strategy: low-voltage analog and digital circuits occupy the left side, and the high-voltage AC monitoring and switching circuits occupy the right side.

A solid ground plane is poured across both top and bottom copper layers on the low-voltage side, providing a low-impedance return path for all analog and digital signals and reducing susceptibility to noise. The high-voltage side has no shared ground plane with the low-voltage domain; electrical isolation is enforced through board-level cutouts and enforced creepage distance between the two regions.

The ACS37800 current path pads (IP+ and IP−) are reinforced with multiple vias stitching the top and bottom copper layers together, increasing the effective copper cross-section and thermal dissipation capacity for the AC current path. High-voltage terminal block pads are similarly reinforced with additional vias and 1.8 mm trace widths routed on both layers stacked directly above each other.

The Raspberry Pi GPIO header is placed at the center of the board, allowing a ribbon cable extension from the rear. Sensor terminal blocks are positioned along the left edge for direct field wiring access, and decoupling capacitors are placed as close as possible to each IC supply pin throughout the board. Four mounting holes at the corners allow the PCB to be securely fastened to standoffs inside the IP65 enclosure, protecting connectors from mechanical stress and ensuring reliable cable connections during long-term operation.

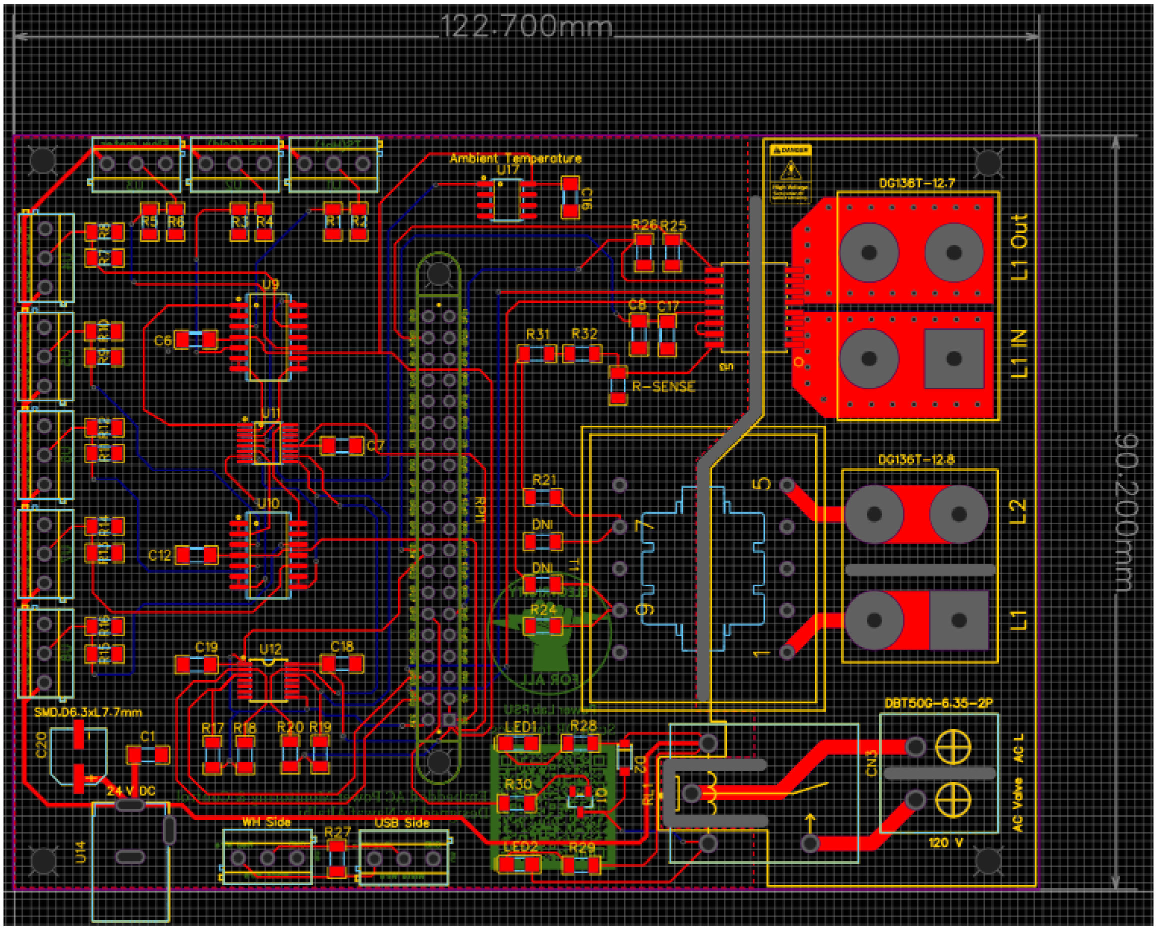

Figure 3: PCB top copper layer (122.7 mm × 90.2 mm). Sensor terminal blocks run along the left edge. The Raspberry Pi 40-pin GPIO header is at the center. High-voltage terminals (L1 IN, L1 OUT, L2, AC valve) occupy the right side. Mounting holes are visible at all four corners.

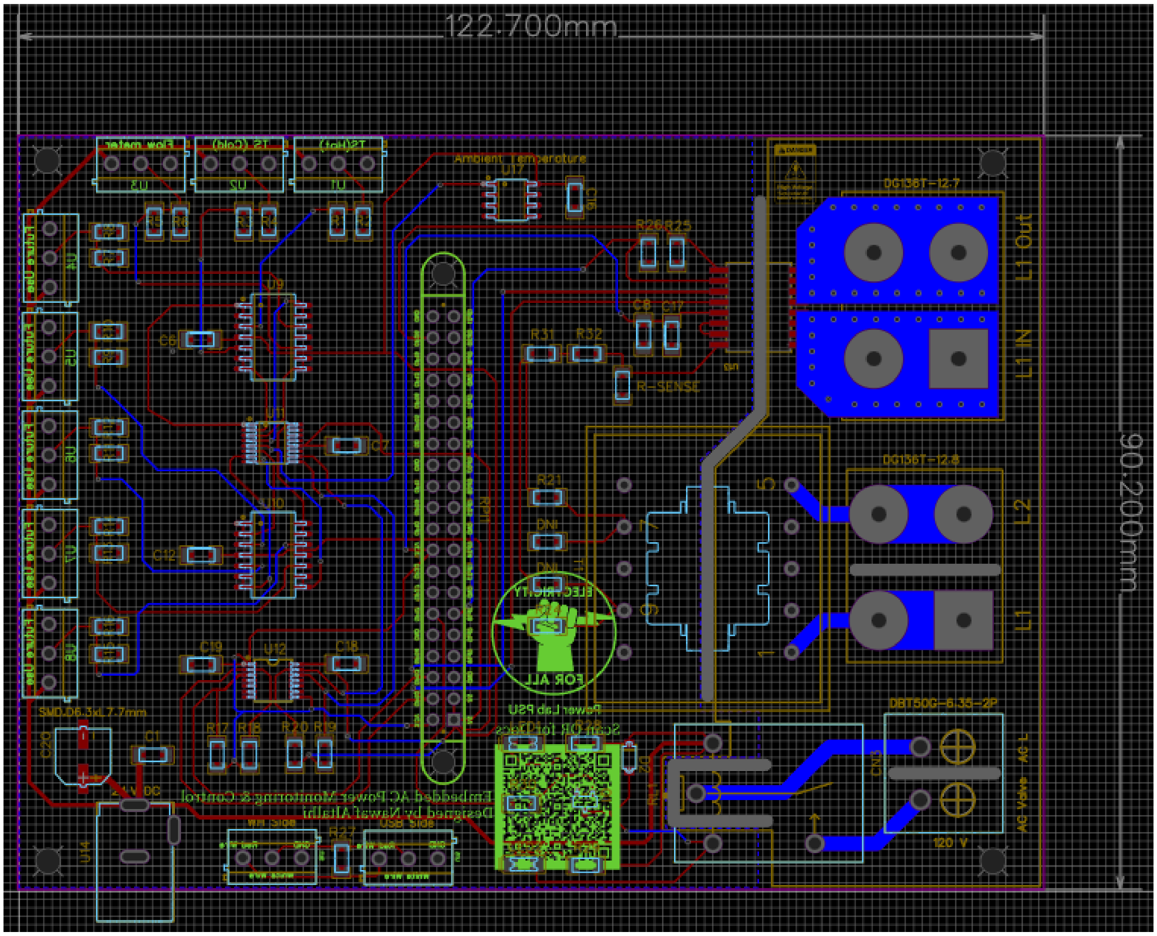

Figure 4: PCB bottom copper layer. Ground plane fills the low-voltage side on both layers. Sensor channel labels and board identification are silkscreened on the rear. A QR code links to the project documentation.

4.2 2D PCB Views

Figures 5 and 6 show the top and bottom 2D views of the PCB as designed in EasyEDA.

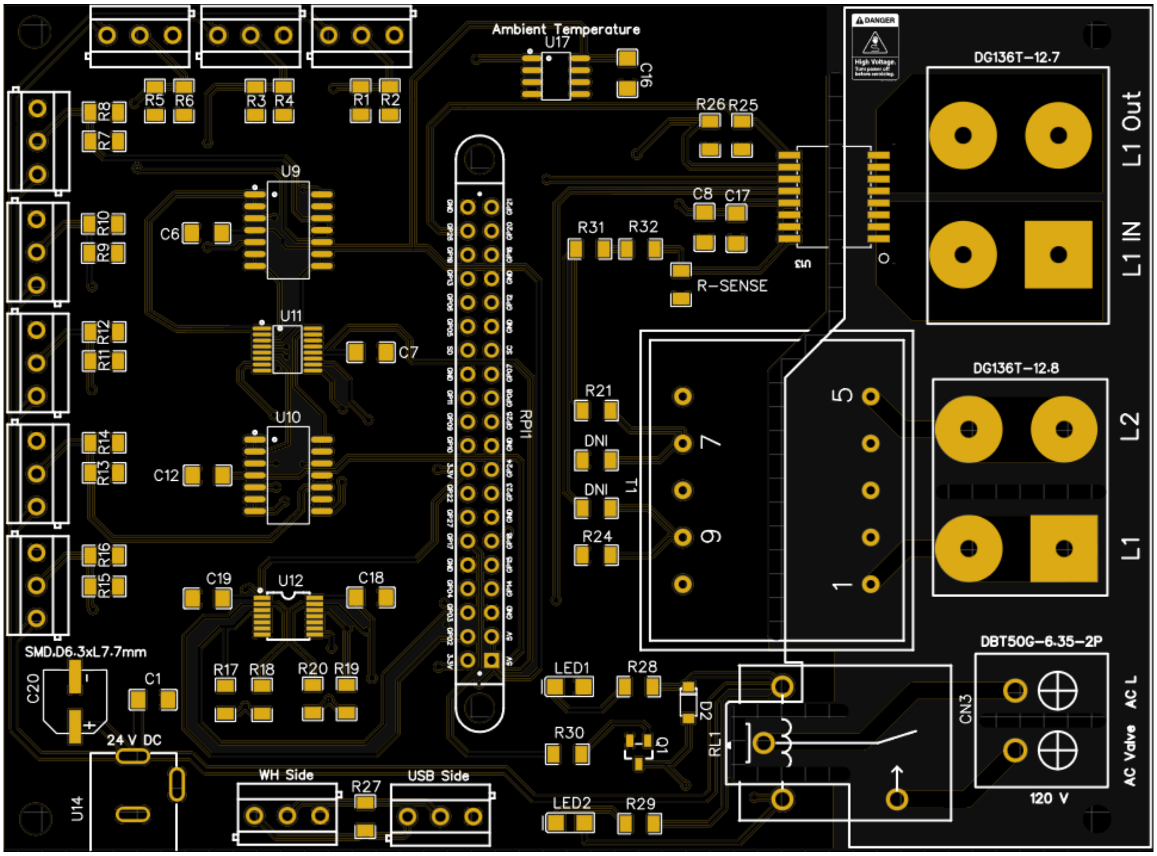

Figure 5: PCB 2D top-side view showing component placement, silkscreen labels, sensor terminal blocks along the left edge, Raspberry Pi GPIO header at center, and high-voltage AC terminals (L1 IN, L1 OUT, L2, AC valve) on the right side. Mounting holes are visible at all four corners.

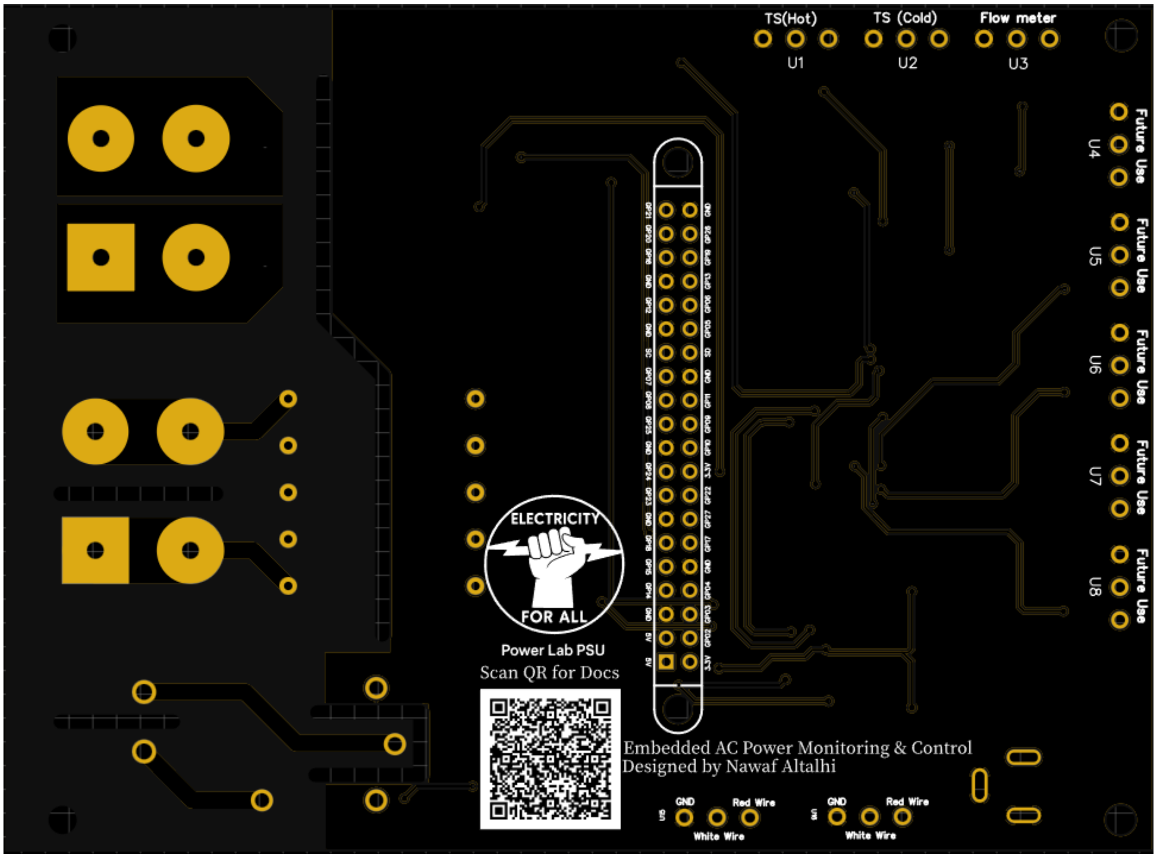

Figure 6: PCB 2D bottom-side view showing rear silkscreen labels including sensor channel identification (TS Hot, TS Cold, Flow Meter, Future Use U4–U8), GPIO pin mapping, wire color reference, QR code linking to project documentation, and board identification: Embedded AC Power Monitoring & Control, designed by Nawaf Altalhi, Power Lab PSU. Mounting holes are visible at all four corners.

4.3 Isolation Strategy

Electrical isolation between the two board regions is enforced at the layout level. The left-side ground plane does not extend into the right-side high-voltage region. Isolation cuts are placed between the transformer primary and secondary, around the ACS37800 current path, and around the relay and AC-valve section. Additional spacing cuts separate the L1 and L2 terminals and the AC valve terminal block from the signal area.

4.4 Trace Widths

Table 4: Trace width specification by net type.

Net Type	Width (mm)
Low-voltage signal traces	0.245
24 VDC sensor supply	0.500
Electrolytic capacitor connection	0.750
High-voltage / high-current traces	1.800

High-voltage and high-current traces are routed on both top and bottom copper layers stacked directly above each other, doubling the effective cross-section. The ACS37800 IP+/IP− pads use thickened copper fills with multiple vias to reinforce the current conduction path.

4.5 Practical Design Features

- Sensor channel labels silkscreened on the PCB rear for field reference.
- A QR code on the PCB links to the system documentation including schematics, software, and troubleshooting guide.
- Board ID, revision, and designer identification on the rear silkscreen.
- Four corner mounting holes for secure standoff installation inside the IP65 enclosure.
- All large terminal blocks selected for the required current and voltage ratings of the AC path.

5 Station Build and Electrical Installation

Safety Note — Read Before Working on the Station

Turning OFF the panel breakers does **NOT** de-energize the solenoid valves. Valve L1 is wired upstream of the breaker and remains live. Always disconnect the panel supply cables for both panels OR turn OFF the main breakers before any wiring work. Verify 0 V at all AC terminals before touching any conductor.

5.1 Enclosure Architecture

The station uses a two-tier enclosure layout designed for separation of high-voltage AC distribution from the per-unit embedded electronics:

- **Main panels (Hammond N1J884, NEMA 1, ×2):** Two steel wall-mount enclosures house the AC power distribution. Each panel contains two Eaton FAZ-B25-2-NA-L 2-pole 25 A miniature circuit breakers mounted on a 35 mm DIN rail — four breakers total, one per water heater. Panel 1 feeds WH1 and WH2; Panel 2 feeds WH3 and WH4. The AC mains input (L1, L2, Neutral, Ground) is landed and distributed inside each panel. All wire connections are made through terminal blocks, not free-wire splices, to support safe maintenance and future modifications.

- **Per-unit enclosures (IP65 ABS, 11 × 7.4 × 5.5 in, ×4):** Each water heater has a dedicated IP65-rated clear-cover ABS enclosure. Each box contains one custom PCB, one Raspberry Pi, and four 60 mm USB cooling fans for thermal management. Cable grommet entries handle AC lines from the breaker panels, 24 VDC power supply, and sensor cables from the water heater pipe assembly.

Figure 7: Station overview showing the two Hammond main panels housing the breakers, four per-unit IP65 enclosures containing the custom PCBs and Raspberry Pi controllers, and the monitoring display connected to a Raspberry Pi for real-time data visualization and system control.

5.2 AC Power Distribution and Wiring

Figure 8 shows the complete AC power wiring diagram for the station.

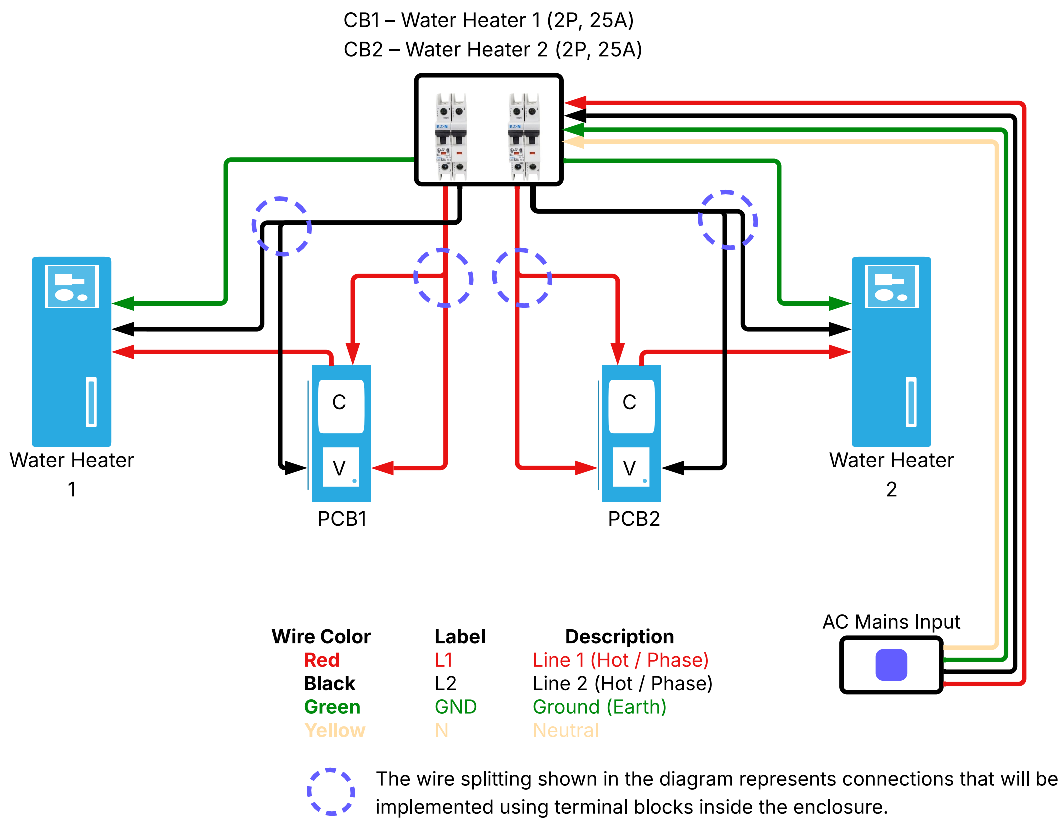

Figure 8: AC power wiring diagram for the Water Heater Test Station. AC mains (L1, L2, Neutral, Ground) are distributed across two Hammond panels through four Eaton 2-pole 25 A breakers — one per water heater. The diagram represents the 240 VAC configuration (L1 and L2); WH1, WH3, and WH4 follow this wiring. WH2 is the exception: it is currently configured for 120 VAC, using L1 and Neutral in place of L1 and L2 — the Neutral wire replaces L2 at the breaker output and at the PCB AC input terminals. WH3 and WH4 are wired identically to WH1. Wire splits inside each panel use terminal blocks. Each PCB taps its supply lines for voltage sensing and routes L1 through the ACS37800 current path for energy monitoring.

Table 5: AC wire color identification used in the station.

Color	Function	Notes
Red	L1 (hot)	From panel breaker; through ACS37800 current path
Black	L2 (hot)	240 VAC return (WH1, WH3, WH4); not used on WH2
Yellow	Neutral (N)	Used on WH2 (120 VAC configuration) instead of L2
Green	Ground (PE)	Protective earth; always connected

Voltage configuration note: The station is designed for 240 VAC (L1 + L2). WH2 is currently configured for 120 VAC using L1 + Neutral. If WH2 is reconfigured to 240 VAC, replace the Neutral wire with L2 and update the power-monitoring calibration constants.

5.3 Pipe Order and Plumbing Installation

The pipe order for each water heater unit, from the water heater outlet to the sink, is:

Figure 9: Pipe assembly for one water heater unit showing all installed devices in sequence: hot water temperature transmitter, 120 VAC solenoid valve, manual flow-adjustment valve, and flow transmitter, connected using NPT fittings and brass reducing bushings.

Figure 10: Overview of the four water heater units installed in the Power Engineering Lab. Each unit is paired with its own pipe assembly, sensor set, and per-unit IP65 control enclosure housing the custom PCB and Raspberry Pi.

- Temperature transmitters (TWTADE, 4–20 mA) are installed using 1/2 in NPT male to 1/4 in NPT female brass reducing bushings.
- The cold water temperature transmitter is on the shared common cold supply. Only one PCB can read it at a time; reassign via terminal block or reuse the value in software.
- The flow transmitter (McMaster-Carr 4386N107) uses a 3/4 in NPT female connection.

- The manual lever-handle valve (McMaster-Carr 4682K55) is installed in series with the flow transmitter for repeatable flow control during testing.

5.4 Wiring and Connection Reference

Complete connector pin assignments, AC terminal identification, RS-485 wiring, status LED behavior, low-voltage connections, and the step-by-step power-down procedure are documented in the dedicated wiring and connection guide produced for this station:

Reference Document

PCB Wiring & Connection Guide — Doc WCG-1, Rev A, June 14, 2026

This document is printed at A3 landscape and **mounted at the station** for use by lab technicians and future students. It is also available in the Power Lab documentation repository (GitHub / Google Drive).

Always refer to WCG-1 before performing any wiring work or opening any enclosure.

6 Calculations

6.1 Current-to-Voltage Conversion

$$R_{eq} = \frac{240 \times 240}{240 + 240} = 120\ \Omega \tag{1}$$

$$V_{4\,\mathrm{mA}} = 0.004 \times 120 = 0.48\ \mathrm{V} \tag{2}$$

$$V_{20\,\mathrm{mA}} = 0.020 \times 120 = 2.40\ \mathrm{V} \tag{3}$$

Quick field check: measure 0.48 V to 2.40 V across the 120 Ω resistor for any active 4–20 mA channel to verify sensor loop integrity.

6.2 ADC Resolution and Code Range

$$\text{Resolution} = \frac{5\ \mathrm{V}}{2^{12}} = \frac{5}{4096} \approx 1.22\ \mathrm{mV/bit} \tag{4}$$

$$\mathrm{Code}_{4\,\mathrm{mA}} = \frac{0.48}{5} \times 4095 \approx 393 \tag{5}$$

$$\mathrm{Code}_{20\,\mathrm{mA}} = \frac{2.40}{5} \times 4095 \approx 1966 \tag{6}$$

6.3 Recovering Sensor Current from ADC Reading

$$I\ [\mathrm{mA}] = \frac{V_{\mathrm{ADC}}}{120\ \Omega} \tag{7}$$

The Python logging script uses this relation to convert ADC codes back to sensor current, then applies the sensor transfer function to obtain the engineering unit (°C or GPM).

6.4 LM35 Output

$$V_{\mathrm{LM35}} = T\ [°\mathrm{C}] \times 10\ \mathrm{mV}/°\mathrm{C} \tag{8}$$

The LM35 is read directly by the ADC on channel AIN4. No op-amp stage is required.

6.5 ACS37800 Voltage Sensing Divider

No-load secondary voltage measured: $V_{sec} \approx 20$ VAC.

$$R_{total} = 1\,\mathrm{M\Omega} + 1\,\mathrm{M\Omega} + 13\,\mathrm{k\Omega} = 2{,}013\,\mathrm{k\Omega} \tag{9}$$

$$V_{INP} = 20 \times \frac{13{,}000}{2{,}013{,}000} \approx 0.129\ \mathrm{V} \tag{10}$$

This is safely below the ACS37800 VINP absolute maximum of 250 mV peak, confirming safe operation of the voltage sensing circuit.

6.6 Software Calibration Constants

Scale factors and offsets are applied in the Python logging script to align raw ACS37800 register values with calibrated reference instruments (multimeter for VRMS, clamp meter for IRMS). Calibration reference: 240.0 V line, 18.8 A.

Table 6: ACS37800 software calibration constants.

Constant	Value
`vrms_scale`	8.7748×10^{-3}
`irms_scale`	1.6478×10^{-3}
`vrms_offset`	168
`irms_offset`	16

$$V_{RMS} = (\text{raw} - \texttt{vrms\_offset}) \times \texttt{vrms\_scale} \tag{11}$$

$$I_{RMS} = (\text{raw} - \texttt{irms\_offset}) \times \texttt{irms\_scale} \tag{12}$$

Recalibrate whenever the power supply path, current-sensing path, or reference instruments change.

7 Testing and Validation

7.1 Sensor Validation

With 24 VDC applied, the output voltage across each 120 Ω sense resistor was measured and confirmed within the expected range. Table 7 shows measured versus expected values.

Table 7: Sensor validation — measured voltage versus expected range.

Condition	Expected (V)	Measured (V)	Result
4 mA sensor output	0.48	0.48	Pass
20 mA sensor output	2.40	2.39	Pass

ADC readings over I^2C were consistent with the applied sensor output on all active channels.

7.2 ADC and I^2C Communication

All active ADC channels (hot temperature, flow, LM35) were read back over I^2C from the Raspberry Pi. Readings were stable with no bus errors. The TXS0104 level shifter operated correctly at the 3.3 V/5 V interface.

7.3 Relay and Valve Control

GPIO17 was toggled by a Python test script. Relay switching was confirmed audibly and by continuity check at connector CN3. LED2 (green) illuminated on valve-open; LED2 extinguished on valve-closed. LED1 (red) remained on with 24 VDC present. The solenoid valve responded correctly with no abnormal behavior.

7.4 Power Monitoring Validation

The ACS37800 was validated under real load during an Electric Mode test running from Wednesday 6:30 PM to Thursday 12:30 PM — an 18-hour continuous test. A controlled water draw event was performed at 10:47 PM (Event 2) to trigger a second heater activation cycle. Reference measurements were taken simultaneously using a calibrated multimeter (VRMS) and a clamp meter (IRMS). After applying the calibration constants from Table 6, the ACS37800 readings were in close agreement with the reference instruments. Table 8 summarizes the key measured parameters.

Table 8: ACS37800 Electric Mode test results — key measured parameters.

Parameter	**Reference**	**ACS37800 (measured)**	**Result**
VRMS (max)	240.0 V	244.21 V	Pass
IRMS (max)	18.8 A	19.45 A	Pass
Power (max)	—	4678.75 W	Pass
Power factor	—	1.000	Pass

Two heater activation events were observed. Event 1 occurred at test start when the heater ran for the first time. Event 2 occurred at 10:47 PM after the water draw, triggering a second heating cycle. A small voltage drop was observed at each heater activation, confirming the accuracy of the ACS37800 voltage measurement. The power factor reached 1.000, indicating efficient resistive heating operation.

Figures 11–13 show the full test results and zoomed views of both events.

7.5 Full System Integration

All four units were operated simultaneously. Water flow was controlled through the solenoid valves, sensor data was collected across all channels, power monitoring ran on each unit, and the Python logging script captured data to local CSV files without errors. The system operated as designed under lab conditions.

7.6 Troubleshooting

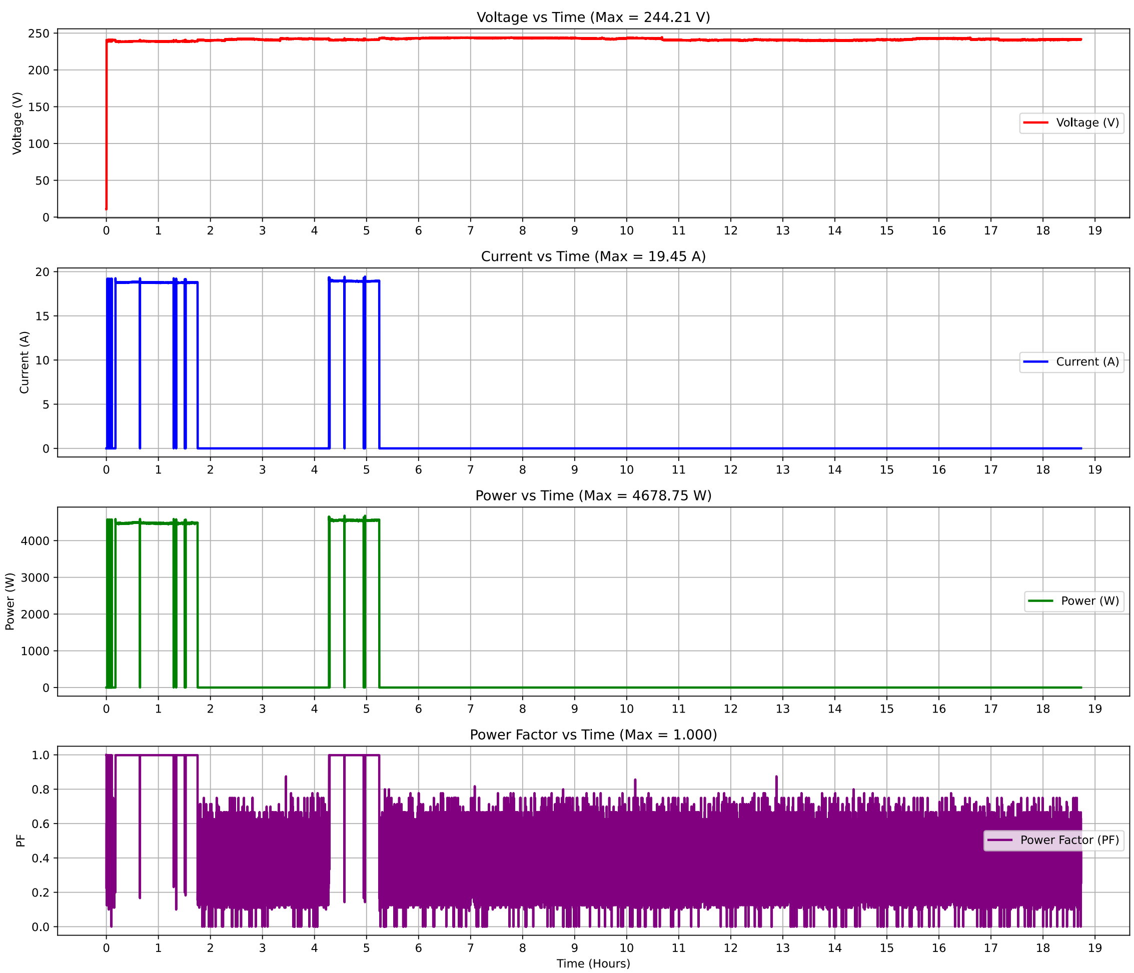

Figure 11: Full Electric Mode test (18 hours) showing VRMS, IRMS, Power, and Power Factor over the entire test duration. Two heater activation events are visible. Maximum values: VRMS = 244.21 V, IRMS = 19.45 A, Power = 4678.75 W, PF = 1.000.

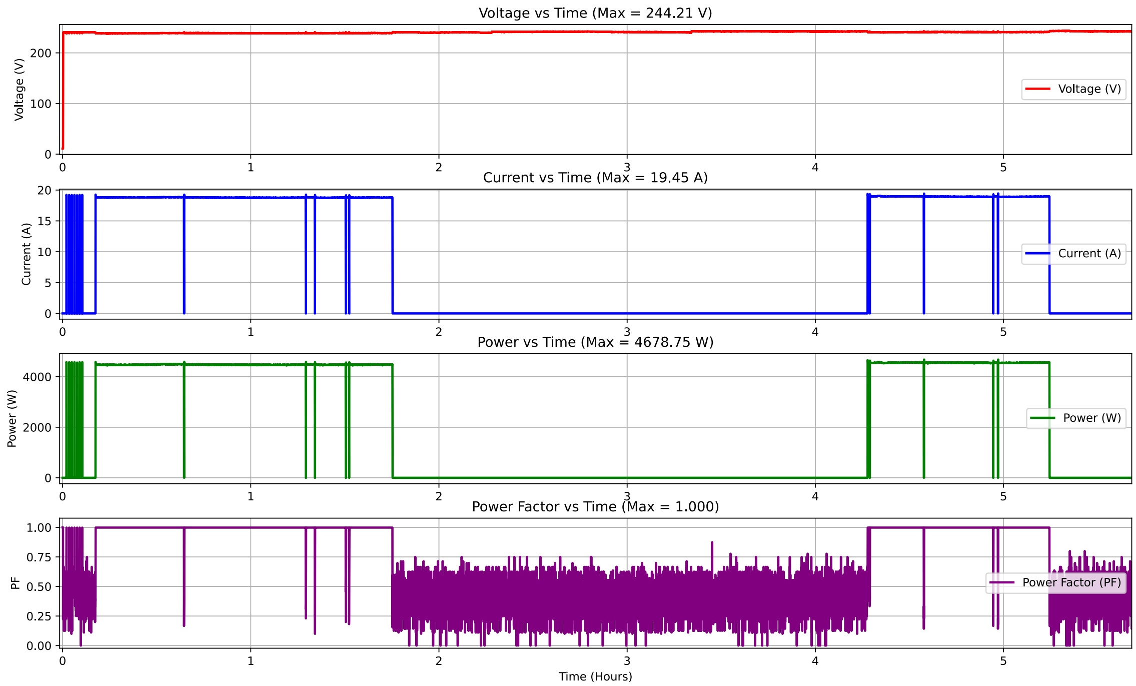

Figure 12: Zoomed view of Event 1 showing the first heater startup at the beginning of the test. Current and power rise sharply when the heating element energizes, with voltage showing a slight drop confirming load detection.

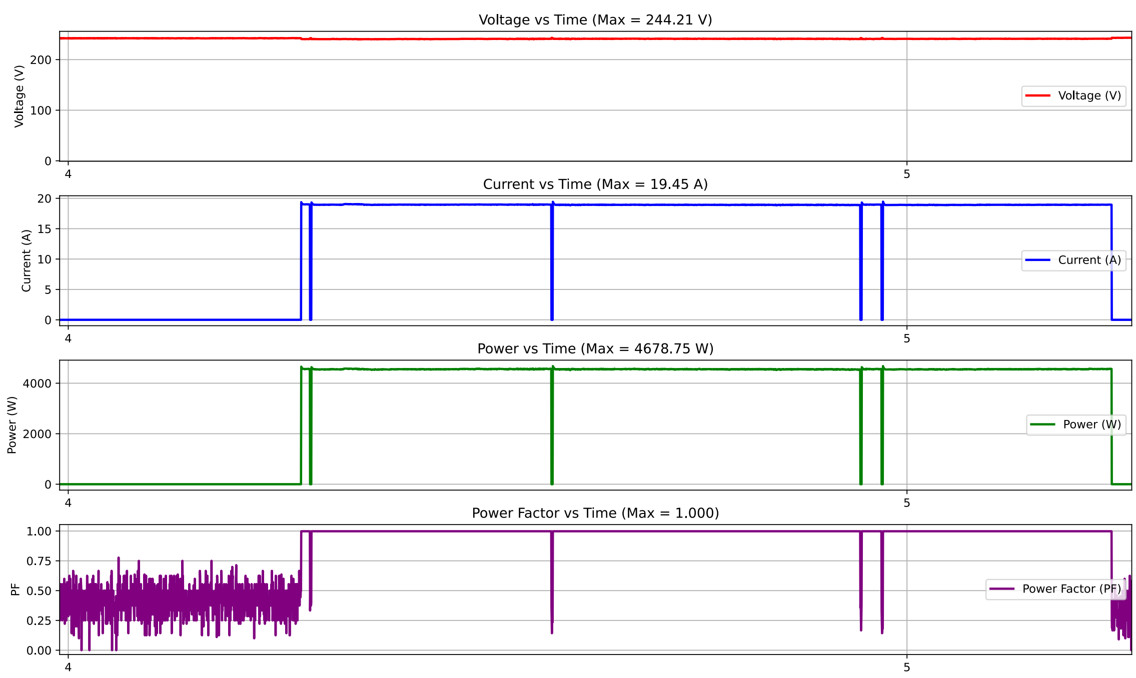

Figure 13: Zoomed view of Event 2 showing heater activation after the controlled water draw at 10:47 PM. The heater responds correctly to the temperature drop caused by the draw, confirming end-to-end system behavior.

Table 9: Common issues and first steps for resolution.

Symptom	Likely Cause	First Step
LED1 (red) is OFF	24 VDC adapter not connected or failed	Check barrel jack connection and adapter output voltage
LED2 (green) does not turn ON	GPIO17 not asserted; relay not energized	Toggle GPIO17 in Python and check relay click audibly
ADC not responding over I^{2}C	Level shifter not powered or address conflict	Verify 5 V and 3.3 V rails are present; run `i2cdetect`
Sensor reads 0 V or stuck	Open loop in 4–20 mA wiring	Measure voltage across 120 Ω resistor; check terminal connections
ACS37800 not responding	3.3 V rail absent; I^{2}C address wrong	Check Raspberry Pi 3.3 V pin; verify I^{2}C address with `i2cdetect`
VRMS reads 0	Transformer primary not connected; CN1 open	Check L1/L2 connections at CN1; verify breaker is ON
IRMS reads 0 when heater is ON	Current path interrupted at IP+/IP−	Inspect ACS37800 current path connections on PCB
Solenoid valve does not open	Valve wiring or 120 VAC supply absent	Check CN3 wiring and confirm 120 VAC is present at valve terminals
Python script crashes on startup	I^{2}C bus error or missing library	Check I^{2}C is enabled in Raspberry Pi config; reinstall dependencies
Readings unstable or noisy	Ground loop or loose terminal	Check all GND terminal connections; tighten sensor terminal screws

7.7 Notes for Future Operators

- The cold water temperature transmitter is shared. Only one unit can read it directly at a time. Reassign via terminal block or share the value in software.
- WH3 and WH4 flow transmitter cables use alternate wire colors (Black/White/Red instead of Blue/White/Brown). Always verify by pin number, not wire color.
- WH2 is currently configured for 120 VAC. If changing to 240 VAC: replace Neutral with L2, verify all wiring, and recalibrate.
- Recalibrate the ACS37800 constants whenever the supply or current path changes.
- The Python logging script uses a configuration file for channel mapping and calibration constants. Verify these before starting a new test.
- Valve L1 is live upstream of the panel breaker at all times. Follow the power-down procedure in Doc WCG-1 before working on any enclosure.

8 Conclusion

The Smart Water Heater Test Station is a fully integrated embedded platform for automated testing and evaluation of residential water heaters in a smart-grid research environment. This project covered the complete hardware lifecycle: system architecture definition, component selection, circuit and PCB design, fabrication, electrical panel build, AC distribution and breaker installation, sensor and plumbing integration, embedded software, and system-level validation across all four units.

The station is operational and ready for continued use in smart-grid research and long-term water heater evaluation. A wall-mounted wiring and connection guide (Doc WCG-1, Rev A) is installed at the station for use by lab technicians and future students.

Table 10: Final system status — June 2026.

Item	Status
System architecture and component selection	Complete
Hardware and circuit design (schematic)	Complete
PCB layout design, fabrication, and assembly (×4)	Complete
Main panel wiring and breaker installation	Complete
Per-unit IP65 enclosures built and installed (×4)	Complete
Sensor and plumbing installation	Complete
ADC I^2C communication verified	Complete
Relay control and solenoid valve operation confirmed	Complete
ACS37800 power monitoring tested and calibrated	Complete
Python data-logging script operational on all units	Complete
Full system integration test (all 4 units)	Complete
Wall-mounted wiring and connection guide (WCG-1 Rev A)	Complete

Recommended future work:

- **Unified 24 VDC power supply with on-board regulation.** Currently the Raspberry Pi requires its own 5 V USB-C adapter per unit. A future PCB revision should integrate a DC–DC buck converter (e.g. LM2596 or MP2307) stepping 24 VDC down to a stable 5 V rail, allowing a single 24 VDC adapter to power both the sensor interface and the Raspberry Pi. This eliminates one cable entry per enclosure and simplifies power management across all four units.
- **Touchscreen HMI mounted on the enclosure lid.** Replace the external HDMI monitor with a small touchscreen (e.g. 3.5 in or 5 in Raspberry Pi-compatible DSI display) mounted directly on the IP65 enclosure lid. This eliminates the external screen and keyboard, makes the unit fully self-contained, and allows operators to view live sensor and power data without a separate computer.
- **Web-based monitoring dashboard.** Implement a lightweight web server on each Raspberry Pi (e.g. Flask or Node-RED) serving a real-time dashboard of sensor readings, power parameters, and valve state. This would allow remote monitoring from any device on the lab network without physical access to the enclosures.

A Document Revision History

Table 11: Document revision history.

Revision	Date	Author	Description
A	June 14, 2026	Nawaf Altalhi	Initial release

Future revisions should increment the letter (B, C, . . .), update this table, and update the document header and title page accordingly. Any wiring changes must also be reflected in Doc WCG-1 with a matching new revision.

B Bill of Materials

The bill of materials is divided into two parts: the PCB and electronics installed per unit (Table 12), and the station-wide infrastructure including in-pipe field devices such as sensors, the manual flow-adjustment valve, transmitters, breakers, and enclosures (Table 13). Within each table, components are listed in order of functional importance to the system. Items procured from Amazon are grouped at the bottom of each table for purchasing-channel separation. Several items were ordered as kits or multi-piece packages; their individual sub-counts are not tracked separately and are flagged in the notes following each table.

B.1 PCB and Electronics (per unit, ×4 units total)

Table 12: PCB and electronics components installed on each per-unit board. Ordered by functional importance; Amazon-sourced general items listed last.

Description	Supplier	Part No.	Qty/Unit
MAX1238 12-bit I^{2}C ADC, 4.5–5.5 V	Mouser	700-MAX1238EEE+	1
ACS37800KMACTR-090B3 Power/Energy Sensor	DigiKey	620-ACS37800…-I2CCT-ND	1
LM324 Quad Op-Amp, 14-SO	DigiKey	296-9543-1-ND	2
LM35 Analog Temp Sensor, 0–100 °C, 8-SOIC	DigiKey	LM35DMX/NOPBCT-ND	1
TXS0104EQPWRQ1 I^{2}C Level Shifter	DigiKey	296-41034-1-ND	1
BV302S12015 Transformer, 1.5 VA, 230 V primary	DigiKey	3388-BV302S12015-ND	1
G5LE-1A4 DC24 Relay, SPST 10 A, 24 V coil	DigiKey	Z2254-ND	1
MMBT3904 NPN Transistor, 40 V, SOT-23	DigiKey	4878-MMBT3904CT-ND	1
1N4007W Diode, 1000 V 1 A, SOD-123FL	DigiKey	5272-1N4007WCT-ND	1

Description	Supplier	Part No.	Qty/Unit
MOV-14D431K Varistor, 430 V, 15.5 mm disc[(a)]	DigiKey	P7267-ND	5
RES 240 Ω 1% 1/4 W 1206	DigiKey	311-240FRCT-ND	16
100 µF Electrolytic Cap, 50 V, SMD	DigiKey	PCE5024CT-ND	1
DC Barrel Jack, BKZ Locking Series	DigiKey	SC3832-ND	1
GPIO Pin Header, 40-pos, 2.54 mm	DigiKey	S9175-ND	1
Conn Barrier Strip, 2-circuit, 0.25 in	DigiKey	3PCV-02-006-ND	1
Term Blk 2P Side Entry 12.7 mm PCB	DigiKey	277-9071-ND	2
Screw Terminal Block, Right Angle PCB	DigiKey	1849-1340-ND	2
Amazon-sourced (kit and general items)			
SMD 1206 Resistor/Capacitor Kit[(b)]	Amazon	B0B2ZSXLV9	1 kit
Sensor Terminal Block (external field wiring)	Amazon	B07RJQDT2X	1

Notes for Table 12:

(a) The MOV-14D431K varistor is *not* installed on the populated board. Five units were ordered as spares so the protection device can be added at the AC input if needed in the future.

(b) The SMD 1206 kit supplies the 0.1 µF ceramic decoupling capacitors and miscellaneous small passives used on the board. The kit is counted as a single line item; individual sub-quantities drawn from it are not tracked.

B.2 Station Infrastructure and In-Pipe Field Devices

Table 13: Station-wide hardware including circuit protection, enclosures, in-pipe transmitters, the manual flow-adjustment valve, sensor cabling, power adapters, and accessories. Ordered by functional importance; Amazon-sourced items listed last.

Description	Supplier	Part No.	Qty
Eaton FAZ-B25-2-NA-L 25 A 2-pole MCB, DIN rail	AutomationDirect	FAZ-B25-2-NA-L	4
Hammond N1J884 NEMA 1 Panel, $8 \times 8 \times 4$ in	AutomationDirect	N1J884	2
Variable-Area Flow Transmitter 4–20 mA, 0.2–10 GPM, 3/4 in NPT	McMaster-Carr	4386N107	4
Flow-Adjustment Valve with Lever Handle, 3/4 in NPT	McMaster-Carr	4682K55	4
TWTADE Temp Transmitter 4–20 mA, −50 to +150 °C, 1/4 in NPT	Amazon	B07RKLWBYC	3
TWTADE Industrial Temp Sensor 4–20 mA, 1/2 in NPT, Probe 30 mm	Amazon	B07RFG17VW	2
24 VDC 1.5 A Power Adapter, barrel jack	Amazon	B0DF7GKK44	4
IP65 ABS Vented Enclosure, $11 \times 7.4 \times 5.5$ in	Amazon	B0D25HV4X7	4
M12 4-pin IP67 Waterproof Sensor Cable, 5 m, A-code	Amazon	B0DNX6ZPZQ	4
1/2 in NPT Male × 1/4 in NPT Female Brass Bushing[c]	Amazon	B07SHS78WD	1 pk
USB 60 mm 5 V Cooling Fan[d]	Amazon	B0CKDG6S5W	1 pk

Notes for Table 13:

[c] Brass bushing reducer is sold as a 5-piece pack (1 200 PSI rated). One pack covers all station adapter needs; individual pieces are not tracked separately.

[d] USB cooling fan is sold as a 4-pack. One pack supplies all four control enclosures; individual fans are not tracked separately.

End of Document

This document completes the Rev A hardware design, station build, and system integration documentation for the Smart Water Heater Test Station. For source files, fabrication outputs, software, schematics, test data, and future updates, refer to the project documentation repository.